EXISTENCE OF A U-ADMISSIBLE BLOWUP ON WHICH A PERFECT MODULE OF TOR DIMENSION AT MOST ONE VANISHING ON U ACQUIRES TWO CANONICAL FILTRATIONS WITH PRESCRIBED INVERTIBLE FITTING DIVISORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0ESU (sources/stacks-flat/pinned-original-source.tex line 13266).

Allowed standard prerequisites (editorial).

- Fitting ideals of a finitely presented module and their behaviour under blowing up two ideals. Divisors, Lemma labelled divisors-lemma-blowing-up-two-ideals (Stacks tag 080A), invoked at proof lines 13301-13302. Supplies that after blowing up the product of the Fitting ideals they become invertible; the definition of the blowup used and the reduction to the invertible case are carried out in the counted span.
- Dropping a trivial summand from a presentation of a module. Algebra, Lemma labelled algebra-lemma-add-trivial-complex (Stacks tag 00MT), invoked at proof line 13338. Supplies that a presentation whose matrix has entries generating the unit ideal can be replaced by one with n decreased by one, which is the induction step on n; the construction of the divisors and of the two filtrations stays in the counted span.

Source reference. Original lemma 080A: sources/stacks-chow/tags.txt, source lines 10175–10175. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13301 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 00MT: sources/stacks-chow/tags.txt, source lines 835–835. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13338 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a scheme. Let $\mathcal{F}$ be a perfect $\mathcal{O}_X$-module of tor dimension ≤ 1. Let $U \subset X$ be an open such that $\mathcal{F}|_U = 0$. Then there is a U-admissible blowup*

$$b : X' \to X$$

such that $\mathcal{F}' = b^*\mathcal{F}$ is equipped with two canonical locally finite filtrations

$$0 = F^0 \subset F^1 \subset F^2 \subset \ldots \subset \mathcal{F}' \quad \text{and} \quad \mathcal{F}' = F_1 \supset F_2 \supset F_3 \supset \ldots \supset 0$$

such that for each $n \geq 1$ there is an effective Cartier divisor $D_n \subset X'$ with the property that

$$F^i/F^{i-1} \quad \text{and} \quad F_i/F_{i+1}$$

are finite locally free of rank i on D_i.

Proof. Choose an affine open $V \subset X$ such that there exists a presentation

$$0 \to \mathcal{O}_V^{\oplus n} \xrightarrow{A} \mathcal{O}_V^{\oplus n} \to \mathcal{F} \to 0$$

for some n and some matrix A. The ideal we are going to blowup in is the product of the Fitting ideals $\mathrm{Fit}_k(\mathcal{F})$ for $k \geq 0$. This makes sense because in the affine situation above we see that $\mathrm{Fit}_k(\mathcal{F})|_V = \mathcal{O}_V$ for $k > n$. It is clear that this is a U-admissible blowing up. By Divisors, Lemma 080A we see that on X' the ideals $\mathrm{Fit}_k(\mathcal{F})$ are invertible. Thus we reduce to the case discussed in the next paragraph.

Assume $\mathrm{Fit}_k(\mathcal{F})$ is an invertible ideal for $k \geq 0$. If $E_k \subset X$ is the effective Cartier divisor defined by $\mathrm{Fit}_k(\mathcal{F})$ for $k \geq 0$, then the effective Cartier divisors D_k in the statement of the lemma will satisfy

$$E_k = D_{k+1} + 2D_{k+2} + 3D_{k+3} + \ldots$$

This makes sense as the collection D_k will be locally finite. Moreover, it uniquely determines the effective Cartier divisors D_k hence it suffices to construct D_k locally.

Choose an affine open $V \subset X$ and presentation of $\mathcal{F}|_V$ as above. We will construct the divisors and filtrations by induction on the integer n in the presentation. We set $D_k|_V = \emptyset$ for $k > n$ and we set $D_n|V = E_{n-1}|_V$. After shrinking V we may assume that $\mathrm{Fit}_{n-1}(\mathcal{F})|_V$ is generated by a single nonzerodivisor $f \in \Gamma(V, \mathcal{O}_V)$. Since $\mathrm{Fit}_{n-1}(\mathcal{F})|_V$ is the ideal generated by the entries of A, we see that there is a matrix A' in $\Gamma(V, \mathcal{O}_V)$ such that $A = fA'$. Define $\mathcal{F}'$ on V by the short exact sequence

$$0 \to \mathcal{O}_V^{\oplus n} \xrightarrow{A'} \mathcal{O}_V^{\oplus n} \to \mathcal{F}' \to 0$$

Since the entries of A' generate the unit ideal in $\Gamma(V, \mathcal{O}_V)$ we see that $\mathcal{F}'$ locally on V has a presentation with n decreased by 1, see Algebra, Lemma 00MT. Further note that $f^{n-k}\mathrm{Fit}_k(\mathcal{F}') = \mathrm{Fit}_k(\mathcal{F})|_V$ for $k = 0, \ldots, n$. Hence $\mathrm{Fit}_k(\mathcal{F}')$ is an invertible ideal for all k. We conclude by induction that there exist effective Cartier divisors $D'_k \subset V$ such that $\mathcal{F}'$ has two canonical filtrations as in the statement of the lemma. Then we set $D_k|_V = D'_k$ for $k = 1, \ldots, n-1$. Observe that the equalities between effective Cartier divisors displayed above hold with these choices. Finally, we come to the construction of the filtrations. Namely, we have short exact sequences

$$0 \to \mathcal{O}_{D_n \cap V}^{\oplus n} \to \mathcal{F} \to \mathcal{F}' \to 0 \quad \text{and} \quad 0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{O}_{D_n \cap V}^{\oplus n} \to 0$$

coming from the two factorizations $A = A'f = fA'$ of A. These sequences are canonical because in the first one the submodule is $\mathrm{Ker}(f : \mathcal{F} \to \mathcal{F})$ and in the second one the quotient module is $\mathrm{Coker}(f : \mathcal{F} \to \mathcal{F})$. □